Scaling Laws for Generative Reward Models

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Abstract

We study generative reward models (GenRMs) as online preference providers for policy optimization in creative writing. Across Qwen3 reward-model sizes from 0.6B to 14B, Thinking judges trained with GRPO have higher static Gold-label accuracy than Answer-Only judges trained with SFT. This ranking often reverses when the judges train policies through online DPO: Answer-Only rewarders yield higher peak and top-five-average policy Elo in seven of eight matched settings under a fixed, learned 32B Gold evaluator. Intermediate Thinking-judge checkpoints can also outperform later, more statically accurate checkpoints as rewarders. Sweeps over rewarder size, policy size, training duration and DPO regularization show substantial benefits from reward-model capacity, including judges larger than the policy. We analyze trajectories with evaluator-specific Elo arenas, conditional match-resampling intervals and descriptive scaling fits. These results demonstrate an *evaluator–rewarder gap* in the studied pipelines: fixed-distribution judging accuracy does not determine downstream optimization quality. The comparison changes judge-training objective as well as response format, and the main ranking is conditional on its learned reference rather than a universal human-quality result. A limited follow-up using a different learned judge is inconclusive overall.

1 Introduction

Policies trained via reinforcement learning from human feedback systematically exploit imperfections in their reward models, degrading true quality even as proxy metrics improve [Gao et al., 2022, Rafailov et al., 2024]. Generative reward models (GenRMs) have emerged as a promising response: by formulating evaluation as conditional text generation rather than scalar prediction, GenRMs can leverage pre-training priors and chain-of-thought reasoning and achieve strong performance on static preference benchmarks [Mahan et al., 2024, Zhang et al., 2025, Whitehouse et al., 2025]. This success has fostered an implicit assumption that more accurate static evaluators should translate to more effective online rewarders.

Prior work has already shown that static reward-model accuracy can be a poor predictor of downstream optimization [Wen et al., 2025, Frick et al., 2024, Kim et al., 2025]. We study this distinction specifically for generative rewarders as the policy distribution changes, targeting a practitioner-facing question: *if one chooses a GenRM as the reward signal, how should compute and architecture be allocated to maximize downstream policy quality?* We study three design dimensions in the online setting where they matter: (i) thinking versus answer-only judging, (ii) judge scale, and (iii) policy scale. We instantiate these axes in creative writing using LitBench [Fein et al., 2025], a non-verifiable preference domain where correctness cannot be programmatically checked. In such domains, static evaluator accuracy can saturate while reward hacking and

Observation	Evidence in this paper
Static evaluation favors Thinking GenRMs	Thinking judges have higher Gold-label validation accuracy across the studied sizes (Figure 1).
Online rewarder quality is different	In online DPO, Answer-Only judges usually give higher peak Gold Elo and less proxy–Gold drift than Thinking judges; the same pattern holds under top-5 checkpoint averaging (Table 3, Figure 2).
Judge scale matters most reliably	Larger Answer-Only GenRMs improve policy quality even when the judge is much larger than the 4B policy; Thinking judge scale helps but saturates earlier (Figure 4).
Static accuracy can be over-trained for reward use	Intermediate Thinking checkpoints can outperform fully-trained Thinking checkpoints as online rewarders (Figure 2b).

Table 1: A compact map of the paper's central evidence. The main empirical distinction is between GenRMs as *evaluators* on a fixed distribution and GenRMs as *rewarders* inside an online optimization loop.

proxy-Gold divergence remain salient, making creative writing an ideal testbed for whether stronger static evaluators translate to better in-loop reward signals.

Contributions. (1) We present a systematic scaling study of GenRMs deployed as online reward signals, sweeping judge format, judge size (0.6B–14B), policy size (0.6B–8B) and training budgets under online DPO. (2) We introduce a proxy–Gold Elo diagnostic that measures alignment under optimization pressure and supports cross-configuration comparison. (3) We provide empirical guidance for GenRM deployment: judge scale and training duration matter strongly, answer-only GenRMs usually yield more stable optimization than thinking variants, and intermediate GenRM checkpoints can outperform fully-trained models as rewarders.

Our findings reveal an *evaluator–rewarder gap*. Thinking GenRMs have higher static validation accuracy across the studied sizes, yet this ranking often reverses during online optimization. Answer-Only GenRMs usually yield higher peak Gold Elo, with exceptions where Thinking remains competitive. Additional judge capacity remains valuable above policy size; this does not establish that judging is intrinsically harder than generating.

2 Methodology

We begin with a human preference dataset of the form

$$\mathcal{D}_{\text{human}} = \{x^{(i)}, y_A^{(i)}, y_B^{(i)}, I_H^{(i)}\}_{i=1}^{N},$$

where $x^{(i)} \in \mathcal{X}$ are prompts, $y_A^{(i)}, y_B^{(i)} \in \mathcal{Y}$ are pairs of responses to the prompts, and $I_H^{(i)} \in \{A, B\}$ denotes the human-preferred response.

Gold evaluator. Since human preferences are not available on demand during policy training, we cannot directly query them to measure policy performance at each checkpoint. To enable continuous evaluation throughout the optimization process, we train a large generative reward model (GenRM), denoted V_{gold}, on $\mathcal{D}_{\text{human}}$. This model serves as a *Gold* evaluator that provides a stable, queryable preference distribution anchored to human judgments. Grounding V_{gold} in human preferences ensures that our experimental setup reflects real-world alignment scenarios where the ultimate objective is to match human judgment. However, our study does not require V_{gold} to perfectly replicate human preferences. Rather, we require a *consistent anchor distribution* against which we can measure proxy GenRM behavior as policies drift during training. The central question

we investigate is how well smaller proxy GenRMs generalize when used as reward signals, relative to this anchor—not whether the anchor itself is perfectly human-aligned.

To make evaluation computationally feasible, we choose V_{gold} to be an *Answer-Only* model that outputs a single indicator token. Using V_{gold}, we construct a Gold Preference dataset:

$$\mathcal{D}_{\text{gold}} = \{x^{(i)}, y_A^{(i)}, y_B^{(i)}, I_G^{(i)}\}_{i=1}^{M},$$

which is then used to train a variety of smaller GenRMs.

Proxy GenRMs. We consider two types of GenRMs: *Answer-Only* and *Thinking*. The Answer-Only models output a direct judgment token (A or B):

$$I \sim v_{\text{ans}}(\cdot \mid x, y_A, y_B),$$

while the Thinking models first generate a reasoning trace z before producing the final verdict:

$$z \sim v_{\text{think}}(\cdot \mid x, y_A, y_B), \quad I \sim v_{\text{think}}(\cdot \mid x, y_A, y_B, z).$$

Training objectives. We train Answer-Only models using supervised fine-tuning (SFT) and Thinking models using GRPO. This asymmetry reflects the training pipelines used by practitioners in real-world settings. For Answer-Only models, the supervision signal is unambiguous: given a preference label, the model should predict the corresponding token. SFT directly optimizes this objective and is the natural choice. For Thinking models, however, the supervision problem is fundamentally different. A chain-of-thought has no single correct trajectory, and current practice overwhelmingly relies on reinforcement learning objectives like GRPO. Our goal is not to perform an architectural ablation under identical training algorithms, but rather to compare the best-supported pipeline for each model type—training each with the method that empirically performs best for its structure.

For GRPO, we employ two reward signals:

1. **Accuracy reward:** A binary reward (1 if the model reaches the correct verdict, 0 otherwise).
2. **Positional consistency reward r_{pos}:** Inspired by prior work, we observe that models often produce contradictory judgments when the order of (y_A, y_B) is swapped in the prompt. The reward manager links both orderings using a shared pair identifier and computes a majority vote within each ordering's sampled completions. An additional reward of 1 is assigned to correct completions only when both majorities are correct and consistent after normalizing response order. This cross-order reward aggregation is distinct from GRPO advantage normalization, whose groups are keyed by the sampled prompt row.

We refer to these trained GenRMs as *Proxy models*, and evaluate them with respect to V_{gold} preferences.

Policy optimization. We train policies π_θ of varying sizes using *Online Direct Preference Optimization* (Online DPO). DPO is a natural choice for preference-based policy optimization because GenRMs produce pairwise preferences as their native output, which DPO can consume directly without requiring scalarization or additional transformations. In *offline* DPO, response pairs are sampled once from a fixed reference policy and reused across many gradient updates, keeping the supervision distribution static throughout training. In *online* DPO, response pairs are repeatedly resampled from the current policy after each update, so the supervision distribution evolves as the policy changes. This online setting is particularly important for our study, as we aim

to understand GenRM robustness under distribution shift induced by policy optimization. Empirically, online preference optimization has been shown to yield more stable and effective alignment dynamics when implemented carefully [Guo et al., 2024].

Formally, at online update t, we sample $x \sim \mathcal{P}_{\text{train}}$ and two responses $y_1, y_2 \sim \pi_{\theta_t}(\cdot \mid x)$. A proxy GenRM v orders the pair as (y_v^+, y_v^-), where y_v^+ is preferred to y_v^-. The DPO loss induced by this proxy is

$$\mathcal{L}_{\text{DPO}}(\theta; v, \pi_{\theta_t}) = -\mathbb{E}\left[\log \sigma\left(\beta\left[\log \frac{\pi_\theta(y_v^+ \mid x)}{\pi_{\text{ref}}(y_v^+ \mid x)} - \log \frac{\pi_\theta(y_v^- \mid x)}{\pi_{\text{ref}}(y_v^- \mid x)}\right]\right)\right].$$

The idealized Gold-aligned update replaces v with V_{gold}. One way to characterize rewarder reliability is to compare its induced update with the Gold-aligned update,

$$B_t(v) = \left\|\nabla_\theta \mathcal{L}_{\text{DPO}}(\theta_t; v, \pi_{\theta_t}) - \nabla_\theta \mathcal{L}_{\text{DPO}}(\theta_t; V_{\text{gold}}, \pi_{\theta_t})\right\|.$$

We do not estimate $B_t(v)$ directly; it clarifies why static GenRM accuracy need not determine online policy quality. Appendix A.6 makes this distinction precise. On a fixed Gold-oriented pair, flipping the preference label changes the DPO gradient by exactly $\beta\nabla_\theta\Delta_\theta$, where Δ_θ is the policy/reference log-probability margin. Thus, online rewarder quality depends on where the proxy is wrong under the policy-induced pair distribution and how large those pairwise update features are. Because the trainer also drops invalid, tied, and position-inconsistent proxy verdicts, a proxy can fail either by mislabeling retained pairs or by selecting a different retained-pair distribution. Static agreement on $\mathcal{D}_{\text{gold}}$ measures neither effect directly.

The policies are trained on prompts sampled from the same broad creative-writing source distribution as the Gold and Proxy datasets, but with LitBench prompts excluded to avoid overlap. During training, we periodically save checkpoints and sample responses on a fixed validation set of prompts. To evaluate these checkpoints, we compute ELO ratings based on pairwise comparisons of their responses. ELO evaluation provides a more fine-grained measurement than raw win rates against a fixed reference distribution. We compute ELOs with respect to both Proxy models and the Gold model, enabling us to analyze the relationship between Proxy-based evaluation and Gold-standard evaluation. Appendix A.2 formalizes this estimator as Bradley–Terry inference from partial comparisons, and Appendix A.5 separates static preference-estimator convergence from the online coverage condition needed for rewarder reliability.

This separates two quantities that are often conflated. Static evaluator agreement is

$$A_{\text{static}}(v) = \Pr_{(x, y_A, y_B) \sim \mathcal{D}_{\text{gold}}}\left[v(x, y_A, y_B) = V_{\text{gold}}(x, y_A, y_B)\right],$$

whereas online agreement at checkpoint t is

$$A_{\text{online}}(v, \pi_{\theta_t}) = \Pr_{\substack{x \sim \mathcal{P}_{\text{val}} \\ y_A, y_B \sim \pi_{\theta_t}(\cdot|x)}}\left[v(x, y_A, y_B) = V_{\text{gold}}(x, y_A, y_B)\right].$$

If $F_T(v)$ denotes the policy obtained after T online-DPO updates using proxy v, and $J_{\text{gold}}(\pi)$ denotes Gold-evaluated policy quality, the evaluator–rewarder gap is the failure of the implication

$$A_{\text{static}}(v_1) > A_{\text{static}}(v_2) \;\not\Rightarrow\; J_{\text{gold}}(F_T(v_1)) > J_{\text{gold}}(F_T(v_2)).$$

Our experiments test this implication directly for Answer-Only and Thinking GenRMs.

3 Experimental Setup

Dataset construction. Our experimental design separates human supervision, Gold-derived supervision and policy-generation prompts:

- **Human preference dataset $\mathcal{D}_{\text{human}}$:** We sample 22,000 preference triplets (x, y_A, y_B, I_H) from LitBench Fein et al. [2025], a large-scale preference dataset over human-written stories from Reddit. We split these into 21,000 for training and 1,000 for validation. This dataset supplies human supervision for the Gold evaluator.

- **Gold preference dataset $\mathcal{D}_{\text{gold}}$:** We sample a separate, non-overlapping set of 22,000 triplets from LitBench and re-annotate them using the trained Gold evaluator, yielding (x, y_A, y_B, I_G) triplets. While both datasets are sourced from LitBench, the former reflects human judgments and the latter reflects the Gold model's judgments. The intended split is 21,000 for training and 1,000 for validation, before filtering invalid or inconsistent Gold labels. This dataset is used for training and evaluating Proxy GenRMs.

- **Policy training dataset:** We source prompts from the WritingPrompts dataset[1], the ancestor dataset from which LitBench was constructed. To avoid data leakage, we exclude all prompts that appear in LitBench, yielding 199,000 candidate prompts. We randomly sample 91,000 prompts (90,000 for training and 1,000 for validation) unless specified otherwise in specific experiments.

Building on prior work that has extensively evaluated various reward models on LitBench facilitates future research and cross-comparisons between our results and other studies.

Model families. We use the Qwen3 model family for all GenRMs and policies. Qwen3 provides both Thinking and Answer-Only modes across multiple parameter sizes, enabling controlled comparisons where we can vary GenRM mode and scale while holding the backbone architecture and instruction-tuning distribution fixed. For policy models, using the same backbone as the GenRMs allows us to cleanly isolate the interaction between policy size and GenRM size, which is central to our scaling analysis.

Gold GenRM training. We train the Gold evaluator from Qwen3-32B using supervised fine-tuning on $\mathcal{D}_{\text{human}}$. To improve positional robustness, we train on both orderings (y_A, y_B) and (y_B, y_A), resulting in 42,000 ordered training examples before any input filtering. We use a batch size of 128 and a learning rate of 1×10^{-5}.

Proxy GenRM training. We train Proxy GenRMs from the Qwen3 series at sizes {0.6B, 1.7B, 4B, 8B, 14B} using $\mathcal{D}_{\text{gold}}$. We refer to off-the-shelf models as `Baseline-[size]-Ans` and `Baseline-[size]-Think`. Answer-Only models use supervised fine-tuning with batch size 128 and learning rate 1×10^{-5}. Both positional orders are used, giving up to 42,000 ordered examples before Gold-label filtering. Thinking models use GRPO with 16 prompts per step, 8 completions per prompt, configured minibatch size 64 and learning rate 1×10^{-6}. The retained launcher disables the trainer-level KL reward controller but enables a separate actor KL loss with coefficient 3×10^{-5}; this is not a zero-KL setup. We refer to trained proxies as `GenRM-[size]-Ans` and `GenRM-[size]-Think`.

Policy training. We train policies π_θ from the Qwen3 series at sizes {0.6B, 1.7B, 4B, 8B} using Online DPO with thinking disabled. We use a prompt batch size of 192, global minibatch size of 64 preference pairs, learning rate 1×10^{-6} and $\beta = 0.02$ unless varied. Two current-policy

[1] `https://huggingface.co/datasets/euclaise/WritingPrompts_preferences`

Axis	Values swept	Default / anchor
Proxy GenRM format	Answer-Only SFT, Thinking GRPO	Both compared
Proxy GenRM size	0.6B, 1.7B, 4B, 8B, 14B	4B unless varied
Policy size	0.6B, 1.7B, 4B, 8B	4B unless varied
Online DPO β	0.1, 0.02, 0.01, 0.005, 0.002	0.02
Training trajectory	Policy evaluation every 10 steps; intermediate Thinking GenRM checkpoints	About 460 policy steps

Table 2: Axes studied in matched and fixed-axis online-DPO sweeps, not a full factorial grid. The design separates fixed-distribution accuracy from in-loop rewarder behavior.

responses are generated per prompt. The proxy judges both orderings, and the trainer sums order-normalized verdicts, discarding zero-sum pairs. A malformed verdict contributes zero, so one valid verdict and one malformed verdict can still yield a retained pair; this is not a guarantee of two valid agreeing votes. The original experiment description specifies prompt and response caps of 512 and 2048 tokens, respectively. The retained hand-edited launcher instead contains caps of 500 and 1500; resolved per-run configurations have not been recovered for every plotted trajectory, so those original caps should be read as nominal settings rather than a newly verified run-wide invariant.

Evaluation. During policy training, we save validation generations every 10 training steps; the retained launcher saves model weights every 50 steps. We call the evaluated policy states evaluation snapshots, without implying that every state has separately retained weights. We compute Elo ratings from pairwise comparisons judged by the proxy and Gold evaluators. Each evaluator has its own fitted ratings over the relevant contestants and comparisons; axes from different evaluators are not one shared numerical scale. Main experiments run for about 460 steps, giving 47 evaluation snapshots including step zero; ablations can differ.

Artifact reconstruction. Dataset counts above follow the contemporaneous experiment account and retained construction code. They describe raw splits, not an independently reconstructed census of every consumed post-filtering example. Some original filtering and Gold-training-label preparation steps, and the complete per-run configuration ledger, are not retained in the inspected source. We distinguish these provenance limits from the stored generations, rating tables and figures used in the analysis.

Reproducibility and LLM usage. Beyond the GenRM and policy models described above, we did not use LLMs for dataset filtering, rubric creation, label consolidation, or prompt generation. All prompts used for policy generation and GenRM pairwise evaluation are provided in Appendix B.

4 Experiments and Results

4.1 GenRM Training

Gold evaluator alignment. The Gold GenRM achieves 79% agreement with human preferences on the held-out validation split of $\mathcal{D}_{\text{human}}$. Prior work on LitBench reports peak accuracies of approximately 78% Fein et al. [2025], indicating that our Gold evaluator achieves strong alignment with the human preference distribution. While our study only requires a stable reference distribution rather than perfect human alignment, the 79% agreement demonstrates that the Gold evaluator is well-grounded in human judgments.

Trained GenRMs substantially outperform baselines. Both Answer-Only SFT and Thinking GRPO improve static Gold-label accuracy over their off-the-shelf counterparts (Fig-

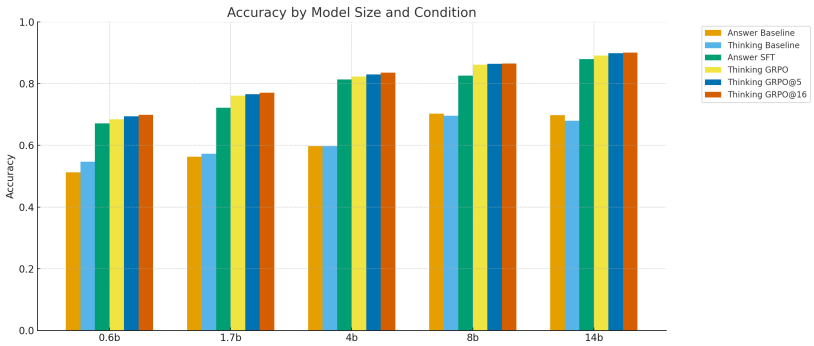


Figure 1: Static evaluator accuracy on the Gold Preference dataset. Thinking GenRMs exceed Answer-Only models across the studied sizes, while both trained variants substantially exceed off-the-shelf baselines. The size of the static advantage varies by model.

ure 1). Similar baseline accuracies do not rule out evaluator-specific preferences or format bias. The trained-versus-baseline improvement is substantial across the sizes studied.

GRPO is an effective training method for Thinking GenRMs. To validate our choice of GRPO for training Thinking models, we conducted an ablation study on the 4B model. When evaluated on the Gold Preference dataset, the off-the-shelf model achieves 60.0%, STaR achieves 60.7%, distillation from QWEN3-32B thinking traces (SFT-style) achieves 64.7%, and GRPO with accuracy and positional consistency rewards achieves 83.2%. Only GRPO yields substantial improvements, validating its use as the primary training method for Thinking GenRMs.

Thinking GenRMs outperform Answer-Only models on in-distribution evaluation. Thinking models have higher static Gold-label accuracy at each studied size (Figure 1); the gap varies by size. Majority voting over $k = 5$ and $k = 16$ Thinking completions gives modest further gains. Given the additional computation, we use one completion per judged ordering during policy training.

4.2 Trained GenRMs vs. Baseline Models in Policy Training

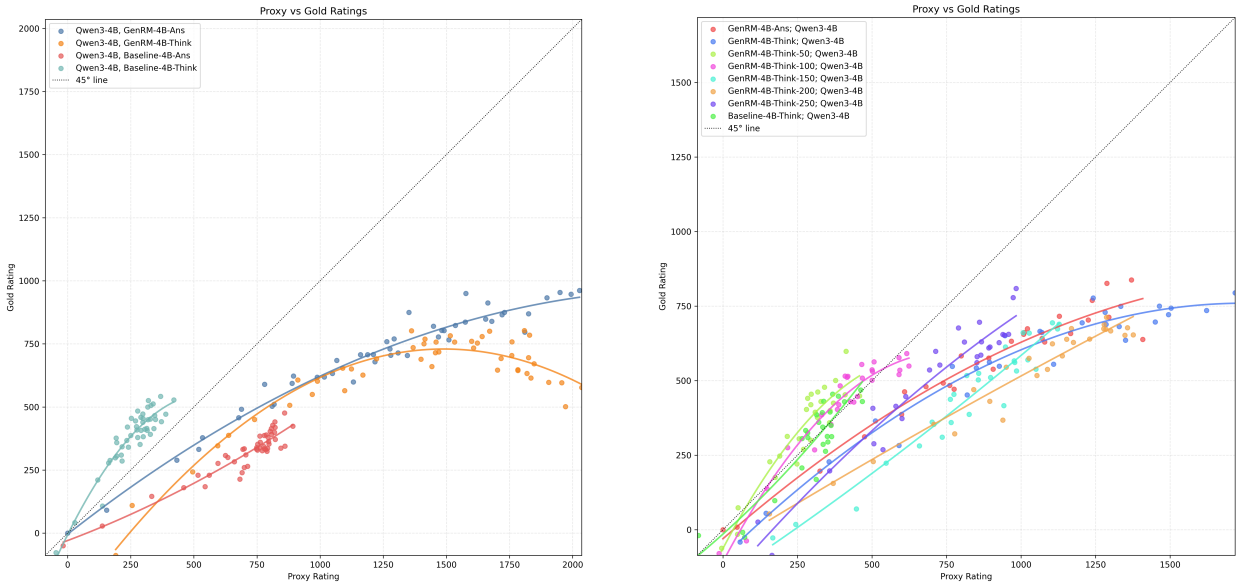
RESEARCH QUESTION. Does training the GenRM improve downstream policy quality compared to off-the-shelf models?

We compare policy training using trained versus baseline (off-the-shelf) GenRMs. In this experiment, both the policy and GenRM are 4B models, with Answer-Only and Thinking variants (Figure 2a). Trained GenRMs dramatically outperform their baseline counterparts. Notably, while Thinking outperforms Answer-Only at baseline, this advantage reverses after training: `GenRM-4B-Ans` exceeds `GenRM-4B-Think` by over 200 Elo points. We discuss this reversal further in the next subsection.

4.3 Answer-Only vs Thinking

RESEARCH QUESTION. Does chain-of-thought reasoning improve GenRM effectiveness as online rewarders?

In the shared global Gold arena, Answer-Only rewarders give higher peak policy Elo in 7 of 8 matched configurations (Table 3). The exception is the 0.6B policy with a 4B judge: Thinking peaks at 1048 versus 1013 for Answer-Only. Averaging each run's five highest-rated snapshots gives the same 7-of-8 pattern, with mean gaps of 107 Elo for top-five averages and 126 for peaks. Under



(a) Trained vs. baseline GenRMs

(b) Intermediate Thinking checkpoints

Figure 2: Proxy vs. Gold ELO ratings for policies trained with different GenRMs. (a) Comparison of trained vs. baseline GenRMs. (b) Policies trained with intermediate Thinking GenRM checkpoints. Axes show separately fitted Proxy and Gold Elo; the dotted diagonal marks equal numerical ratings, not identical verdicts. **Takeaway:** The evaluator-best checkpoint need not be the rewarder-best. The original experiment account specifies 40,000 training prompts for (b).

200 independent-row bootstrap resamples and regularized refits, 5 of 8 peak gaps and 5 of 8 top-five gaps have 95% percentile intervals above zero. This is conditional match-table uncertainty, not prompt-cluster, generation or retraining uncertainty (Appendix A.2). Figure 3 preserves the full trajectories and competitive small-scale or early-training Thinking settings. The reversal matters because static evaluation favors Thinking, and the off-the-shelf baseline comparison also differs from the trained comparison. Under the learned Gold reference, the statically stronger practical judge-training pipeline is not generally the better online reward provider. We interpret this as evidence consistent with Thinking GenRMs being more vulnerable to off-distribution shifts introduced by policy training. Two factors may explain this gap.

First, higher in-distribution GenRM accuracy does not necessarily translate to better policy training. Figure 2b compares intermediate 4B Thinking checkpoints at 0, 50, 100, 150, 200, 250 and 300 GRPO steps. The contemporaneous experiment account reports static accuracies of 59.7%, 61.5%, 64.5%, 68.7%, 71.5%, 79.2% and 82.3%, respectively. The 250-step judge (79.2%) produces a stronger online rewarder than the later, more statically accurate checkpoint in this ablation. This illustrates that evaluator-optimal and rewarder-optimal checkpoint selection can differ. This late ablation used 40,000 policy-training prompts rather than the main 90,000.

Second, GRPO supervises final preferences rather than the correctness of each reasoning step. A Thinking GenRM may therefore learn traces that support training-distribution labels without remaining reliable under policy-induced shift; we treat this as a hypothesis, not a measured mechanism. The trajectory data show differing stability: across the eight matched global configurations, Answer-Only runs lose an average of 60 Elo from peak to final, compared with 130 for Thinking. An auxiliary verdict-agreement reconstruction could align rows only by checkpoint-player pairs and row order, not unique prompt/response identifiers. We do not use its percentages as verified

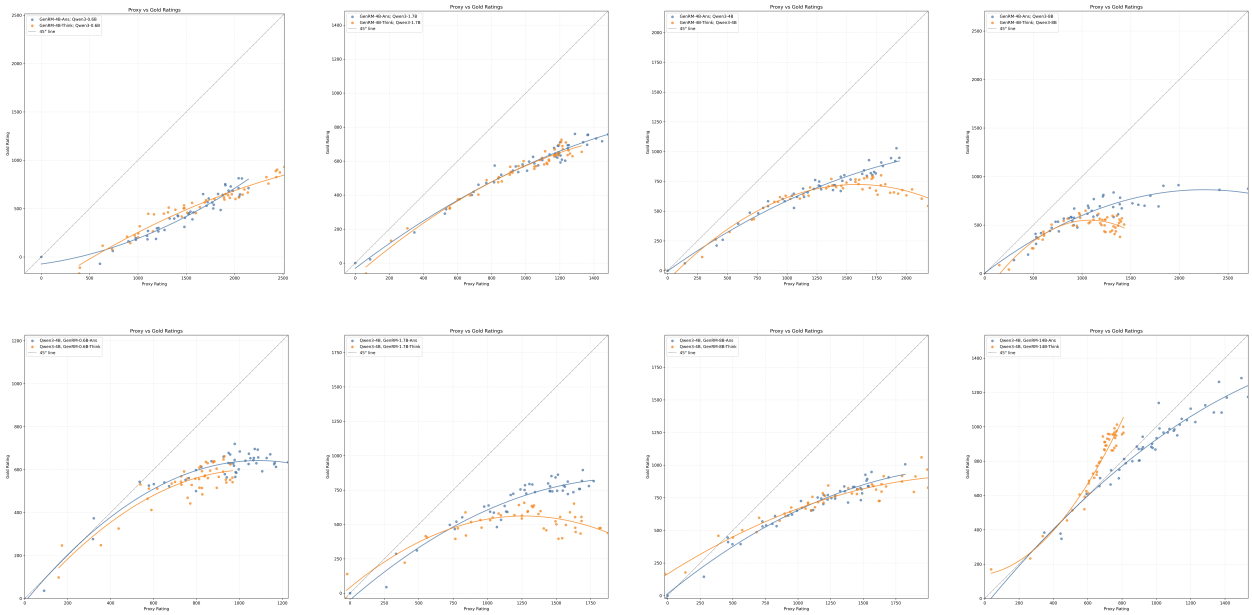


Figure 3: Proxy vs. Gold Elo for Answer-Only (blue) and Thinking rewarders (orange). Top: policy sizes 0.6B, 1.7B, 4B, 8B with a 4B rewarder. Bottom: a 4B policy with 0.6B, 1.7B, 8B, 14B rewarders. The diagonal marks equal numerical ratings in separately fitted arenas. **Takeaway:** Answer-Only usually gives higher Gold Elo; Thinking remains competitive in some settings.

response-level agreement or as mechanistic evidence.

These results should not be read as evidence that Answer-Only supervision preserves stylistic diversity. A surface audit of stored peak-checkpoint generations in two 4B-policy cells shows that high Gold Elo can coexist with template convergence: in the 4B-policy/4B-judge cell, 94% of Answer-Only peak outputs share the same opening line, and the duplicate-line rate is higher than in the corresponding Thinking peak outputs. The stronger Answer-Only result is therefore a preference-optimization result, not a diversity result. In summary, Answer-Only reward models are more token efficient and more robust on average for preference-based policy optimization in this domain, though intermediate Thinking checkpoints retain potential if their training dynamics are better understood.

4.4 Effect of GenRM Size

RESEARCH QUESTION. Does scaling the reward model improve policy quality, and do gains continue when the GenRM exceeds the policy in size?

We investigate the effect of scaling GenRM size while fixing the policy size at 4B. Figure 4 shows results for both Answer-Only and Thinking GenRMs across scales from 0.6B to 14B. We observe clear gains from increasing GenRM size, especially for Answer-Only rewarders. With a 4B policy, the Answer-Only 14B judge is the strongest judge-size configuration in the global arena. Thinking rewarders also benefit substantially from scale, but less monotonically: the 8B Thinking judge slightly exceeds the 14B Thinking judge in peak Gold Elo (1507 vs. 1451) and top-5 average Gold Elo (1488 vs. 1437). Thus additional rewarder capacity can remain useful above policy size, while its benefit depends on the judge's training and online behavior. Parameter counts alone do not establish a general ranking of the difficulty of judging and generating.

Appendices A.3–A.4 cast these trends as theory-guided scaling-law fits over the existing sweep.

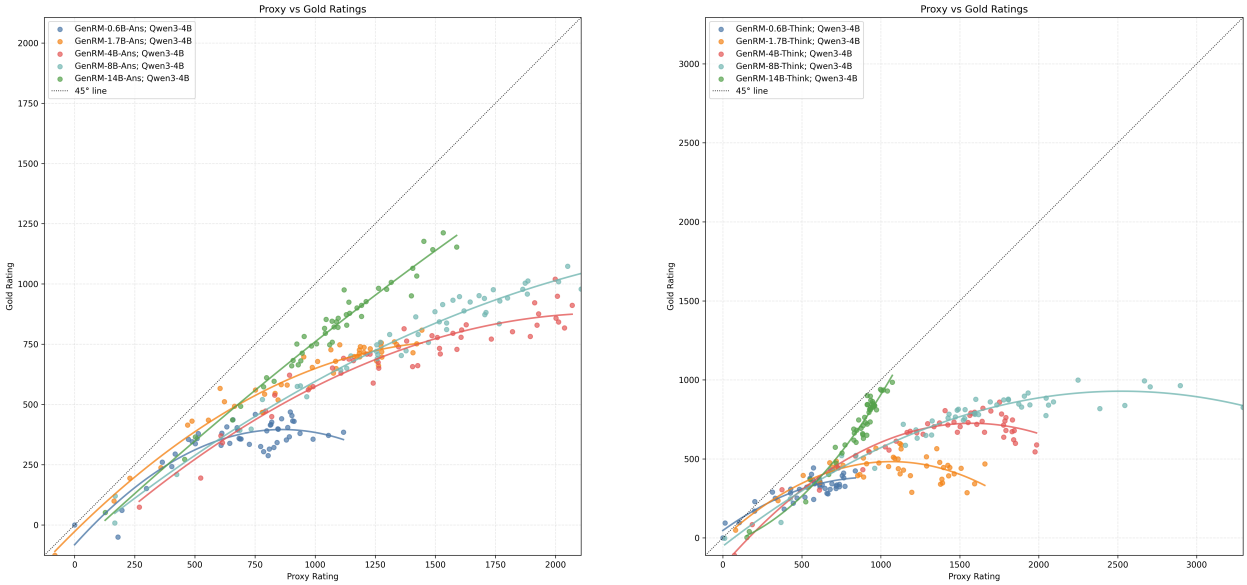
Policy	GenRM	Peak gap [95% CI]	Top-5 gap [95% CI]
0.6B	4B	−35 [−134, 49]	−19 [−59, 25]
1.7B	4B	+113 [43, 176]	+94 [48, 134]
4B	0.6B	+41 [−32, 104]	+24 [−2, 65]
4B	1.7B	+145 [104, 239]	+159 [133, 214]
4B	4B	+135 [49, 241]	+141 [93, 189]
4B	8B	+41 [−67, 143]	+25 [−18, 88]
4B	14B	+269 [128, 488]	+193 [134, 257]
8B	4B	+302 [158, 573]	+243 [189, 343]

Table 3: Matched Answer-Only versus Thinking Gold Elo gaps in the global all-configuration arena, reported as Answer-Only minus Thinking. Confidence intervals come from 200 bootstrap resamples of the stored Gold match table. Answer-Only wins 7 of 8 matched cells under both peak and top-5 checkpoint averaging; the exception is the smallest-policy setting with a 4B judge.

Configuration	Top-5 avg Gold Elo	Peak Gold Elo	Peak step
GenRM-4B-Ans; Policy 8B	1645	1736	450
GenRM-14B-Ans; Policy 4B	1630	1719	430
GenRM-8B-Ans; Policy 4B	1513	1549	460
GenRM-8B-Think; Policy 4B	1488	1507	460
GenRM-4B-Ans; Policy 4B	1467	1488	420
GenRM-14B-Think; Policy 4B	1437	1451	440

Table 4: Top online-DPO configurations in the global all-configuration arena, ranked by average Gold Elo over each run's best five checkpoints. The practical frontier is dominated by larger judges and Answer-Only rewarders, with the 8B Thinking judge remaining competitive.

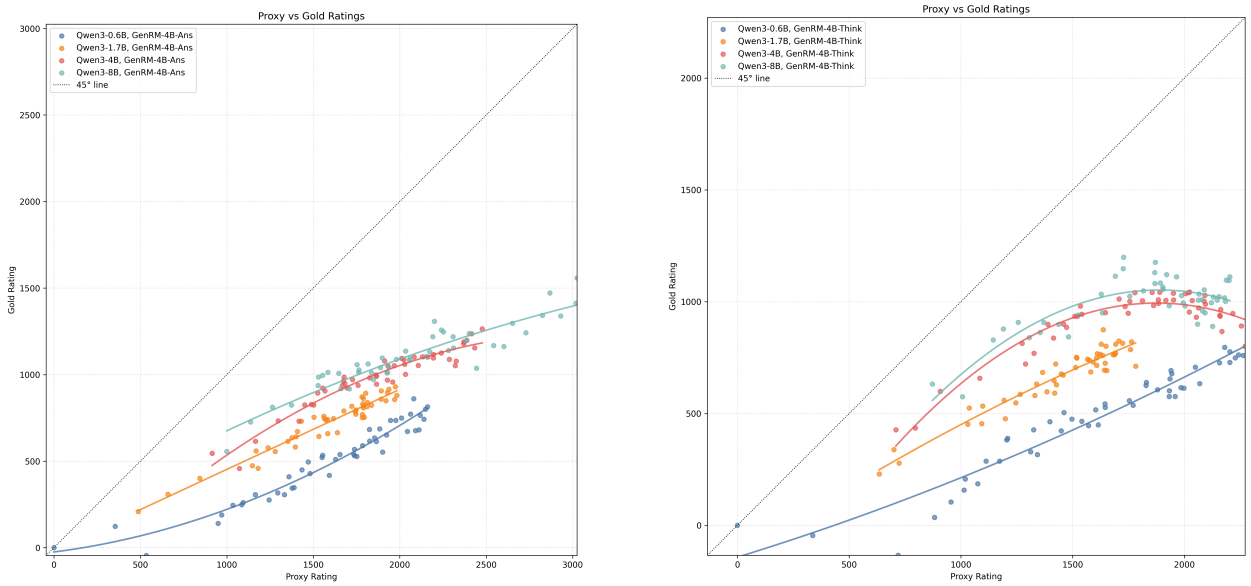
A log-linear model using policy size, judge size, and judge mode explains 95.2% of the variance in top-5 Gold Elo across the 16 matched global configurations. In that fit, an Answer-Only policy-size doubling is worth 176 Elo and an Answer-Only judge-size doubling is worth 136 Elo; for Thinking runs, the corresponding effects are 119 and 116 Elo. Thus the clearest scaling claim is two-dimensional: policy scale and rewarder scale both matter, while the Thinking format changes the effective elasticities and makes the trajectories less monotone.



(a) Answer-Only GenRMs

(b) Thinking GenRMs

Figure 4: Proxy vs. Gold ELO ratings for policies trained with GenRMs of different sizes. The policy is fixed at 4B. Left: Answer-Only GenRMs (0.6B–14B). Right: Thinking GenRMs (0.6B–14B). The diagonal marks equal numerical ratings in separate arenas. **Takeaway:** Larger Answer-Only rewarders improve Gold Elo beyond policy size; Thinking scaling is less monotone.



(a) Answer-Only GenRMs

(b) Thinking GenRMs

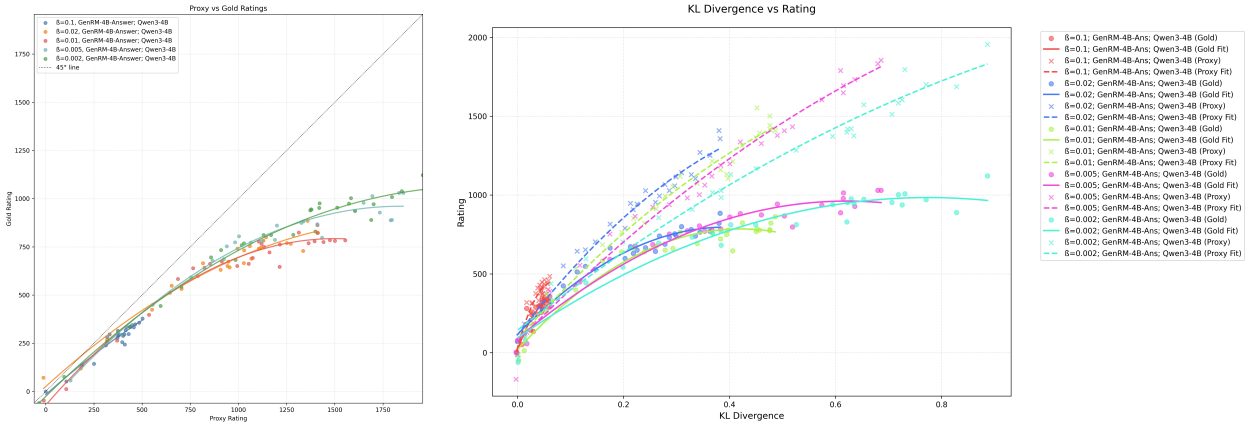
Figure 5: Proxy vs. Gold ELO ratings for policies of different sizes trained with fixed GenRMs. Left: Answer-Only GenRMs. Right: Thinking GenRMs. The diagonal marks equal numerical ratings in separate arenas. **Takeaway:** Larger policies improve under Answer-Only supervision in this sweep; large Thinking-supervised policies saturate earlier.

4.5 Effect on Policy Size

RESEARCH QUESTION. Does scaling the policy improve quality, and does the effect differ between Answer-Only and Thinking supervision?

We examine the effect of scaling policy size while fixing the GenRM. Figure 5 shows results for policies ranging from 0.6B to 8B trained with both Answer-Only and Thinking GenRMs. For Answer-Only supervision, the trend is straightforward: larger policies consistently achieve higher Gold ELO, with steady improvements across scales. This aligns with the expectation that larger policies better exploit the reward signal and generalize more effectively.

For Thinking supervision, however, the picture is less clear. While the largest policy does achieve the highest peak performance, its Gold ELO curve saturates and bends downward earlier than smaller policies, which continue to improve steadily. In the policy-size arena, the peak checkpoint for Thinking supervision moves earlier as policy size increases: the 0.6B and 1.7B policies peak near the end of training, the 4B policy peaks around step 320, and the 8B policy peaks around step 130. This suggests two possible explanations: (i) large policies may more quickly exhaust the effective capacity of the GenRM, reaching its "ceiling" earlier, or (ii) beyond a certain scale, further increasing policy size without correspondingly stronger GenRMs may have diminishing or negative returns. Additional training is required to disentangle these explanations, but the evidence points toward an important asymmetry: scaling policy size is reliably beneficial under Answer-Only supervision, but under Thinking supervision, the interaction between policy capacity and GenRM capacity is more fragile.



(a) Proxy vs. Gold ratings across β values

(b) KL divergence vs. rating

Figure 6: Effect of the β coefficient in Online DPO. Left: Proxy vs. Gold ELO ratings for Answer-Only GenRMs at different β values. Right: KL divergence vs. rating tradeoff, showing how larger β suppresses exploration. The diagonal marks equal numerical ratings in separate arenas. **Takeaway:** Smaller β gives higher Gold Elo among tested values. The original experiment account specifies 40,000 training prompts for this sweep.

4.6 Effect of the β Coefficient in Online DPO

RESEARCH QUESTION. How does the DPO regularization strength affect policy quality and the exploration-exploitation tradeoff?

We analyze the effect of the β coefficient in Online DPO. Figure 6(a) shows Proxy vs. Gold Elo for $\beta \in \{0.1, 0.02, 0.01, 0.005, 0.002\}$. The smallest tested value, $\beta = 0.002$, gives the highest Gold Elo. The ordering is not strictly monotone: 0.02 slightly exceeds 0.01 in the complete stored trajectory. Lower β does not simply increase Proxy ratings at Gold's expense: Gold performance also improves over the observed range. Proximity to the diagonal is a descriptive comparison of separately anchored ratings, not proof of exact proxy–Gold verdict agreement. Figure 6(b) further illustrates the KL–performance tradeoff. Larger β values overly constrain the policy to the reference distribution, limiting attainable Gold ratings. Reducing β relaxes this constraint and improves alignment with Gold outcomes, without inducing severe divergence. The result favors less regularization within the tested range, without identifying an optimum outside it. In the KL-matched plotted subset, peak Gold Elo rises from 357 at $\beta = 0.1$ to 1122 at $\beta = 0.002$, while the corresponding KL estimates rise from 0.061 to 0.887. The complete stored $\beta = 0.1$ trajectory reaches about 378 Elo at step 260, beyond that matched subset. KL values are empirical estimates; small near-zero negative estimates are not interpreted as negative true KL.

5 Limitations and Discussion

Gold evaluator is a learned proxy, not humans. We mitigate label noise by fine-tuning a 32B Gold evaluator on human preferences, re-annotating pairwise data with this model, and discarding items with inconsistent Gold verdicts [Gao et al., 2022]. This improves consistency but couples our objective to the Gold model's inductive biases. At 32B parameters, the Gold judge is stronger than our proxies, leaving headroom for improvement, yet it still reflects preferences from its training set. *Takeaway.* Interpret results as optimization toward a strong, learned proxy.

Evaluator dependence and response quality. Gold uses an Answer-Only Qwen3 model, and proxies share its model family and derived labels. Baseline comparisons do not rule out evaluator-specific preferences. A retained follow-up audit used a different learned judge, `Qwen3-235B-A22B-Instruct-2507`, on 40 randomly selected validation prompts from each of two 4B-policy settings, comparing each mode's peak-Gold outputs with randomized presentation order. Answer-Only won 42 of 80 comparisons (52.5%; 95% Wilson interval 41.7–63.1%): 17/40 with 4B rewarders and 25/40 with 14B rewarders. This limited audit is inconclusive overall and does not independently establish the Gold-ranked advantage. It uses another Qwen3 Answer-Only judge, a different rubric and Gold-selected checkpoints, not human evaluation or a cross-family replication. Appendix A.8 gives the procedure. *Takeaway.* The main result is conditional on the specified learned Gold reference; neither format neutrality nor universal writing-quality superiority is established.

Domain scope. All experiments use creative writing; we do not systematically evaluate other domains such as safety, helpfulness, or verifiable tasks (math, code). Replicating our full experimental grid in additional domains would scale cost roughly linearly: each (judge, policy) configuration requires an online DPO run plus Elo evaluation, costing 6–20 hours on 8×H100, not including judge training. Given our ablation count (policy size × judge size × judge mode × training-budget/β/checkpoint sweeps), full multi-domain replication would require thousands of additional H100-hours. To provide preliminary evidence, we ran cross-domain checks on OpenOrca using off-the-shelf Qwen3 models (Appendix A.1); these support the same qualitative pre-training trend (thinking judges start stronger as static evaluators). *Takeaway.* Our conclusions are scoped to non-verifiable creative writing; domain transfer is an important open question.

Relation to prior creative-writing studies. Creative-writing preferences are subjective. Optimizing win rate can reduce stylistic diversity. We observe some style convergence [Chung et al., 2025] in late-stage policies against both judge modes, while draw rates do not show large collapses. Fein et al. [2025] report that chain-of-thought can degrade verification accuracy for creative writing, and that trained BT and generative verifiers outperform zero-shot judges on their benchmark. In our setting, distilling Qwen3-32B traces into Qwen3-4B also yields negligible gains as evaluators, which aligns with these observations. However, after training, thinking judges significantly improve as static evaluators. The gap between thinking and answer-only during policy optimization therefore cannot be explained by domain "unsuitability" alone; it indicates different optimization dynamics. *Takeaway.* Our evaluation–optimization divergence is a property of the training loop in this domain, not only a property of static judging.

Family and algorithm scope, and behavioral priors. All models use Qwen3 backbones where pretraining data are not public. Cross-family studies suggest that behavioral priors, including synthetic data that instantiate verification or backtracking, can modulate RL improvements and collapse family gaps [Gandhi et al., 2025]. Such priors could shift our coefficients. We do not evaluate PPO-style RLHF for policies or alternative thinking-judge recipes. Additionally, we do not include Best-of-N (BoN) inference-time comparisons; our focus is on GenRMs as *training signals* for policy optimization rather than inference-time scaling, where the primary gains in our setting arise from gradient-based learning rather than reranking at generation time. BoN comparisons remain

valuable future work for understanding the full compute-performance tradeoff. *Takeaway.* Our coefficients and inflection points are conditional on online DPO and Qwen3 underlying behavioral priors.

Elo anchoring and schedule. Ratings depend on evaluator, contestants, anchor and match schedule [Chiang et al., 2024]. Proxy and Gold ratings are fitted separately; the diagonal of a proxy–Gold plot is not a verdict-agreement statistic. We report global and size-stratified arenas. Bootstrap intervals for Answer-Only-minus-Thinking gaps independently resample stored winner/loser rows and refit a lightly regularized Bradley–Terry model. They are conditional on these stored comparisons, not robust to arbitrary dependence between rows sharing prompts or responses, and do not capture fresh generation, retraining, seed, anchor or tournament-design uncertainty. *Takeaway.* Interpret within-arena differences alongside these dependence and selection limits, not as universal confidence statements.

Post-hoc scaling-law fits. The scaling theory in Appendix A.3 derives local signal and curvature terms, but the fitted parameters in Appendix A.4 are estimated from the stored sweep after the experiments were run. They should be read as a compact statistical summary of the observed grid, not as seed-averaged laws or extrapolations to larger models. Most stored KL trajectories do not visibly reach a full downturn region. *Takeaway.* The parameter elasticities and local curvature estimates are useful within this sweep; stronger laws require independent seeds and deliberately extended KL budgets.

6 Related Work

Reward overoptimization is studied in RLHF [Ouyang et al., 2022, Gao et al., 2022] and direct methods such as DPO [Rafailov et al., 2023, 2024]. Prior evaluations explicitly distinguish static accuracy from downstream policy performance [Wen et al., 2025, Frick et al., 2024, Kim et al., 2025]; the generic distinction is not new to this work. GenRMs replace scalar-head prediction with text generation, allowing verdicts and rationales [Mahan et al., 2024, Zhang et al., 2025]. Subsequent approaches include GRPO and positional consistency in J1 [Whitehouse et al., 2025], self-principled critique and aggregation in DeepSeek-GRM [Liu et al., 2025], and test-time majority voting [Wang et al., 2025, Shi and Jin, 2025]. Related supervision uses generated critiques [Yu et al., 2025], criteria trees [Liang et al., 2025] and reasoning plans [Saha et al., 2025]; judge reliability and static benchmarks remain active topics [Ye et al., 2024, Lambert et al., 2024]. Our contribution is the measured set of generative-rewarder capacity and training-pipeline sweeps inside online DPO, using evaluator-specific Elo arenas [Chiang et al., 2024]. The results concern practical Answer-Only SFT and Thinking GRPO pipelines under a shared model family, not a causal isolation of reasoning format.

7 Conclusion

We study generative reward models as online training signals. Under our learned Gold reference, Thinking GenRMs outperform Answer-Only variants on static evaluation, yet this advantage usually reverses during policy optimization. The result concerns the compared SFT/GRPO pipelines in this domain, not an isolated causal effect of reasoning or a universal ranking of writing quality.

Three findings offer practical guidance: (1) reward model scale is a major driver of policy quality, with gains continuing even when the GenRM far exceeds the policy in size; (2) intermediate GenRM checkpoints can outperform fully-trained models as rewarders; and (3) in-distribution accuracy does

not by itself predict training effectiveness. These results suggest that optimizing GenRMs only for static benchmarks can select checkpoints that are suboptimal for downstream alignment.

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A Appendix

Additional experimental details and supplementary results.

A.1 Preliminary Cross-Domain Evaluation: OpenOrca

To provide preliminary evidence on domain generalization (see §5), we ran cross-domain checks on OpenOrca using off-the-shelf Qwen3 policies, judges, and Gold evaluator. No fine-tuning was performed. These results support the same qualitative pre-training trend observed in the main creative-writing domain: prior to GenRM training, Thinking judges show stronger alignment with the Gold evaluator than Answer-Only judges.

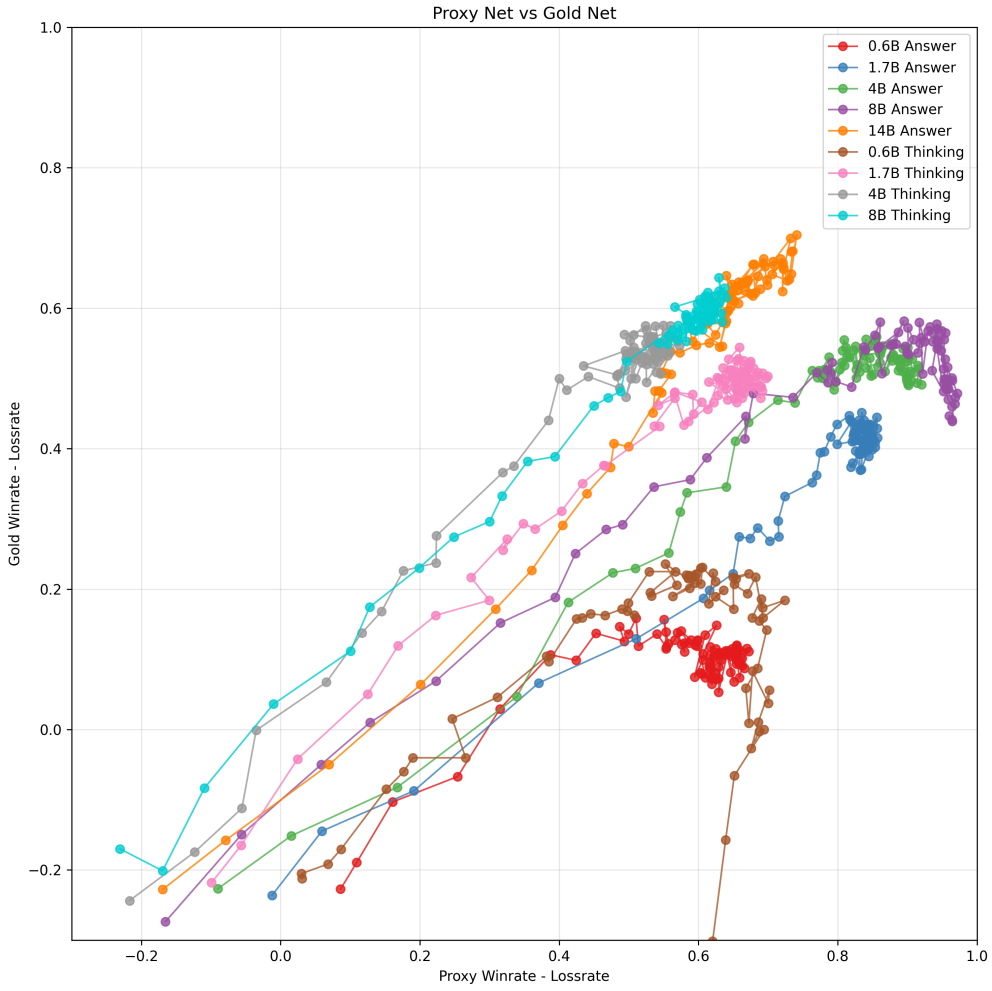


Figure 7: Proxy vs. Gold net score on OpenOrca. All models are off-the-shelf (no GenRM training). A full replication with trained GenRMs and policy optimization remains future work.

A.2 Bradley–Terry Inference from Partial Comparisons

The empirical sections compare many policy checkpoints using pairwise and partial-order preference data. We model these comparisons with a Bradley–Terry likelihood, which is the logistic model underlying Elo-style ratings. Let $\mathcal{S}$ denote the set of checkpoints in an arena, with $K = |\mathcal{S}|$. A comparison record is (i_m, j_m, w_m), where $i_m, j_m \in \mathcal{S}$ and $w_m = 1$ means checkpoint i_m defeats checkpoint j_m under either the Gold evaluator or a proxy GenRM. Each checkpoint has a latent rating $R_i \in \mathbb{R}$, and

$$p_R(i \succ j) = \sigma\left(\frac{R_i - R_j}{s}\right), \qquad \sigma(u) = \frac{1}{1+e^{-u}},$$

where $s > 0$ is a fixed scale. Ratings are identifiable only up to translation, so we anchor one checkpoint s_0 by setting $R_{s_0} = 0$. For a match table $\mathcal{M} = \{(i_m, j_m, w_m)\}_{m=1}^{M}$, the anchored maximum-likelihood estimator is

$$\widehat{R} = \arg\max_{R: R_{s_0}=0} \ell_M(R), \qquad \ell_M(R) = \sum_{m=1}^{M} \left[w_m \log p_R(i_m \succ j_m) + (1 - w_m) \log p_R(j_m \succ i_m)\right].$$

Proxy Elo ratings use a separately fitted model with proxy decisions replacing Gold decisions. The likelihood above describes the ideal unpenalized estimator. For the reported bootstrap analysis, the implemented refit adds an L2 penalty $\frac{0.01}{2}\|\theta_{-s_0}\|^2$ in logit-rating coordinates $\theta = R/s$, with $s = 400/\log 10$. Original exported ratings and the bootstrap's regularized refits are closely related but are not byte-identical estimates.

The likelihood has a useful graph interpretation. Let $b_m = e_{i_m} - e_{j_m}$ be the comparison incidence vector and $B \in \mathbb{R}^{M \times K}$ the matrix with rows $b_m^\top$. Then $p_m(R) = \sigma(b_m^\top R/s)$ and

$$-\nabla^2 \ell_M(R) = \frac{1}{s^2} B^\top W(R) B, \qquad W(R) = \operatorname{diag}(p_m(R)(1 - p_m(R))).$$

Thus the observed information is a weighted graph Laplacian. On the anchored $(K-1)$-dimensional subspace, the objective is strictly concave when the comparison graph connects all checkpoints with nonzero weights. A finite MLE additionally requires no complete separation: equivalently, the directed graph formed by observed wins is strongly connected, so no nonempty subset of checkpoints beats its complement in every cross-cut comparison.

Under the correctly specified Bradley–Terry model, independent comparisons, and a limiting positive-definite Fisher information on the anchored subspace,

$$\sqrt{M}\,(\widehat{R}_{-s_0} - R^*_{-s_0}) \;\Rightarrow\; \mathcal{N}\big(0, \mathcal{I}(R^*)^{-1}\big),$$

where

$$\mathcal{I}(R^*) = \lim_{M \to \infty} \frac{1}{M s^2} B_{-s_0}^\top W(R^*) B_{-s_0}.$$

Equivalently, local uncertainty is controlled by the inverse weighted Laplacian: checkpoints with dense, balanced comparisons have smaller variance, while weakly connected checkpoints have larger variance.

Some evaluation records are partial orders rather than independent binary observations. If an observation o provides a set of implied ordered edges $E(o)$, we use the rank-broken composite likelihood

$$\ell_{\text{comp}}(R) = \sum_{o} \sum_{(i,j) \in E(o)} \log p_R(i \succ j).$$

If edges within the same observation are dependent, the inverse Hessian alone can understate uncertainty; an appropriate asymptotic covariance can require a sandwich form $H^{-1}JH^{-1}$ with dependence-aware score covariance. Our reported bootstrap instead independently resamples stored winner/loser rows, refits the regularized ratings and takes percentile intervals over derived gaps. Prompt and response cluster identifiers were not retained in this table. These intervals therefore do not solve the within-prompt dependence problem; they describe conditional row-resampling sensitivity under an iid-row approximation.

For a configuration c with checkpoint ratings $\{R_{c,t}\}_t$, we report

$$\mathrm{Peak}(c) = \max_t R_{c,t}, \qquad \mathrm{Top5}(c) = \frac{1}{5} \sum_{t \in \mathcal{T}_5(c)} R_{c,t},$$

where $\mathcal{T}_5(c)$ indexes the five highest-rated checkpoints for c. The bootstrap confidence intervals in Table 3 are computed by resampling $\mathcal{M}$ with replacement, refitting $\widehat{R}^{(b)}$ for each bootstrap sample, and taking percentile intervals over the Answer-Only-minus-Thinking gaps in Peak and Top5.

A.3 Population Theory for Proxy-Optimization Scaling Curves

The empirical curves in this paper arise from optimizing a proxy preference model and then evaluating with Gold. We use an idealized cardinal-utility population model to motivate curve shapes. Its identities are exact for the exponential-tilt optimizer defined below, not for arbitrary finite neural online-DPO trajectories. Measured policy Elo summarizes pairwise preferences and need not equal expected cardinal utility. Consider one prompt distribution and a reference policy π_0. Let $U(y)$ be the Gold utility of response y, let $E_R(y)$ be the proxy error at scale R, and define the proxy score

$$V_R(y) = U(y) + E_R(y).$$

The KL-regularized population optimizer for the proxy is the exponential tilt

$$\pi_\lambda(y) = \frac{\pi_0(y) \exp(\lambda V_R(y))}{Z_R(\lambda)}, \qquad Z_R(\lambda) = \mathbb{E}_{\pi_0}\left[e^{\lambda V_R}\right].$$

Here λ is the optimization pressure. Equivalently, π_λ solves

$$\max_{\pi \ll \pi_0} \left\{ \mathbb{E}_\pi[V_R] - \lambda^{-1} D_{\mathrm{KL}}(\pi \| \pi_0) \right\}.$$

Define the Gold curve and KL curve

$$G_R(\lambda) = \mathbb{E}_{\pi_\lambda}[U], \qquad D_R(\lambda) = D_{\mathrm{KL}}(\pi_\lambda \| \pi_0).$$

Proposition 1 (Exact population curve identities)**.** *For every $\lambda > 0$ at which Z_R is finite in an open neighborhood, the relevant moments permit differentiation and $\mathrm{Var}_{\pi_\lambda}(V_R) > 0$,*

$$\frac{dG_R}{d\lambda} = \mathrm{Cov}_{\pi_\lambda}(U, V_R), \qquad \frac{dD_R}{d\lambda} = \lambda \, \mathrm{Var}_{\pi_\lambda}(V_R),$$

and therefore

$$\frac{dG_R}{dD_R} = \frac{\mathrm{Cov}_{\pi_\lambda}(U, V_R)}{\lambda \, \mathrm{Var}_{\pi_\lambda}(V_R)} = \frac{\mathrm{Var}_{\pi_\lambda}(U) + \mathrm{Cov}_{\pi_\lambda}(U, E_R)}{\lambda \, \mathrm{Var}_{\pi_\lambda}(V_R)}.$$

Proof. For any statistic A, differentiating the tilted expectation gives

$$\frac{d}{d\lambda}\mathbb{E}_{\pi_\lambda}[A] = \mathrm{Cov}_{\pi_\lambda}(A, V_R).$$

Taking $A = U$ proves the first identity. Since $D_R(\lambda) = \lambda\mathbb{E}_{\pi_\lambda}[V_R] - \log Z_R(\lambda)$ and $\frac{d}{d\lambda}\log Z_R(\lambda) = \mathbb{E}_{\pi_\lambda}[V_R]$, the derivative of D_R is $\lambda\,\mathrm{Var}_{\pi_\lambda}(V_R)$. Dividing the two derivatives and using $V_R = U + E_R$ gives the final expression. □

Proposition 1 is an exact identity within this KL-regularized population model. Expected Gold utility rises, saturates or falls according to its tilted covariance with the proxy. If $E_R = 0$, the numerator is $\mathrm{Var}_{\pi_\lambda}(U) \geq 0$, so this optimizer cannot produce a downturn in expected Gold utility. A decline therefore requires sufficiently adverse proxy-error covariance under the tilted distribution, not necessarily heavy-tailed errors. An interior stationary point requires

$$\mathrm{Cov}_{\pi_\lambda}(U, E_R) = -\,\mathrm{Var}_{\pi_\lambda}(U),$$

and decline requires the proxy error to become even more anti-correlated with Gold utility under the tilted distribution. Rewarder scaling should therefore be read as changing the whole tilted error geometry, not merely lowering average prediction error on the reference distribution.

Theorem 1 (Local Gold scaling under proxy optimization)**.** *Assume U and V_R have finite exponential moments in a neighborhood of zero under π_0, and let $\sigma_R^2 = \mathrm{Var}_{\pi_0}(V_R) > 0$. Let $\kappa_R^{(3)} = \mathrm{cum}(V_R, V_R, V_R)$ and $\kappa_{U,R}^{(2)} = \mathrm{cum}(U, V_R, V_R)$. Then, for $d = \sqrt{D_R(\lambda)}$ near zero,*

$$G_R(d) - G_R(0) = \alpha_R d + \rho_R d^2 + O(d^3),$$

where

$$\alpha_R = \frac{\sqrt{2}\,\mathrm{Cov}_{\pi_0}(U, V_R)}{\sigma_R}, \qquad \rho_R = \frac{\kappa_{U,R}^{(2)}}{\sigma_R^2} - \frac{2\,\mathrm{Cov}_{\pi_0}(U, V_R)\kappa_R^{(3)}}{3\sigma_R^4}.$$

Proof. Let $K_R(\lambda) = \log Z_R(\lambda)$. The tilted distribution satisfies

$$D_R(\lambda) = \lambda K_R'(\lambda) - K_R(\lambda), \qquad G_R(\lambda) = \frac{\partial}{\partial t}\log \mathbb{E}_{\pi_0}\left[e^{\lambda V_R + tU}\right]\Big|_{t=0}.$$

Cumulant expansion gives

$$D_R(\lambda) = \frac{1}{2}\sigma_R^2\lambda^2 + \frac{1}{3}\kappa_R^{(3)}\lambda^3 + O(\lambda^4)$$

and

$$G_R(\lambda) - G_R(0) = \mathrm{Cov}_{\pi_0}(U, V_R)\lambda + \frac{1}{2}\kappa_{U,R}^{(2)}\lambda^2 + O(\lambda^3).$$

Since

$$d = \frac{\sigma_R}{\sqrt{2}}\lambda + \frac{\kappa_R^{(3)}}{3\sqrt{2}\sigma_R}\lambda^2 + O(\lambda^3),$$

inverting yields

$$\lambda = \frac{\sqrt{2}}{\sigma_R}d - \frac{2\kappa_R^{(3)}}{3\sigma_R^4}d^2 + O(d^3).$$

Substituting into the expansion for $G_R(\lambda)$ gives the result. □

The theorem identifies analogous signal and curvature terms in the population model, not identified coefficients of the empirical Elo fits. The initial slope α_R is a normalized covariance, not simply reward-model accuracy. If proxy error is independent of U with variance τ_R^2, then

$$\alpha_R = \frac{\sqrt{2}\,\mathrm{Var}(U)}{\sqrt{\mathrm{Var}(U) + \tau_R^2}},$$

so reducing reward-model error increases the initial Gold improvement. The curvature ρ_R depends on mixed third cumulants, meaning the downturn or saturation behavior is controlled by tail structure: how proxy score extremes co-vary with Gold utility. This explains why higher static accuracy need not imply a better online rewarder. A Thinking judge can improve average static prediction while still having unfavorable tail/cumulant behavior on policy-induced samples.

Policy restrictions suggest a possible local mechanism. If a policy family can realize only a tangent subspace $\mathcal{T}_P$ near π_0, the useful first-order direction depends on projection into that subspace. More expressive policies may expose both Gold-aligned directions and proxy-error directions. This motivates studying both policy and rewarder capacity; without assumptions linking parameter counts to subspaces and error distributions, it does not derive a numerical parameter-scaling law.

The square-root parameterization follows from the same local geometry. KL is second order in policy displacement: if $\pi_{\theta+\delta}$ is a smooth perturbation of π_θ, then

$$D_{\mathrm{KL}}(\pi_{\theta+\delta} \| \pi_\theta) = \frac{1}{2}\delta^\top F_\theta \delta + o(\|\delta\|^2),$$

where F_θ is the Fisher information. Thus $d = \sqrt{D_{\mathrm{KL}}}$ behaves like a distance along the policy manifold, while raw KL behaves like squared distance.

A.4 Statistical Scaling-Law Fits

The idealized utility model above has a local quadratic in $d = \sqrt{D_{\mathrm{KL}}}$. This motivates, but does not identify, a descriptive quadratic fit to empirical Gold Elo. Prior reward-overoptimization work also uses smooth low-dimensional curves in d [Gao et al., 2022, Rafailov et al., 2024]. We fit two summaries from stored CSV artifacts: a global log-size model and local KL curves. No exact utility-to-Elo or neural-optimizer equivalence is assumed.

First, for each configuration $c \in \mathcal{C}$, let P_c be the policy size in billions, R_c the GenRM size in billions, and $M_c \in \{0, 1\}$ indicate Thinking mode. For $Y_c \in \{\mathrm{Peak}(c), \mathrm{Top5}(c)\}$, we fit the log-elasticity model

$$Y_c = b_0 + b_P \log P_c + b_R \log R_c + b_M M_c + b_{MP} M_c \log P_c + b_{MR} M_c \log R_c + \epsilon_c.$$

The per-doubling effects are $b_P \log 2$ and $b_R \log 2$ for Answer-Only runs, and $(b_P + b_{MP}) \log 2$ and $(b_R + b_{MR}) \log 2$ for Thinking runs. Writing $x_c = (1, \log P_c, \log R_c, M_c, M_c \log P_c, M_c \log R_c)$ and $b = (b_0, b_P, b_R, b_M, b_{MP}, b_{MR})$, the estimator is ordinary least squares:

$$\widehat{b} = \arg\min_b \sum_{c \in \mathcal{C}} \left(Y_c - x_c^\top b\right)^2 = (X^\top X)^\dagger X^\top Y.$$

The log-size form is a local power-law approximation: if quality obeys a smooth power law in policy or rewarder scale over the measured range, then its first-order expansion in log-size is linear. The interaction terms allow Thinking and Answer-Only rewarders to have different elasticities rather than forcing a shared scaling exponent.

Metric	R^2	RMSE	Ans. policy	Ans. judge	Think policy	Think judge
Peak Elo	0.954	51.8	186.0	144.7	113.3	113.3
Top-5 Elo	0.952	51.2	175.8	135.7	119.0	115.6

Table 5: Log-scaling fits over the 16 matched global configurations. Slope columns report Gold-Elo gain per parameter doubling along each axis.

Table 5 quantifies the two-axis nature of the sweep. Answer-Only runs have a larger policy-size elasticity than judge-size elasticity, but judge scaling remains large. Thinking runs have nearly equal policy and judge elasticities, while also exhibiting lower and less monotone gains. This supports the qualitative conclusion from Figures 4 and 5: rewarder capacity is not a small implementation detail, and the mode of the judge changes how useful added capacity becomes.

Second, for each local KL trajectory $\{(D_{ct}, G_{ct})\}_{t=1}^{T_c}$, where G_{ct} is Gold Elo at checkpoint t, Theorem 1 motivates the local quadratic

$$G_c(d) = \gamma_c + a_c d - b_c d^2, \qquad d = \sqrt{\max(D_{\mathrm{KL}}, 0)}.$$

Here a_c and b_c summarize local Elo slope and curvature; they are not identified utility cumulants. A positive interior maximum on $d \geq 0$ requires both $a_c > 0$ and $b_c > 0$, with

$$d_c^* = \frac{a_c}{2b_c}, \qquad G_c(d_c^*) = \gamma_c + \frac{a_c^2}{4b_c}.$$

We also fit the Gao-style direct-alignment comparison curve

$$G_c(d) = \gamma_c + d(\alpha_c - \kappa_c \log d).$$

The intercept γ_c absorbs the local arena anchor. The term $d\alpha_c$ captures the local first-order improvement from moving away from the reference policy, while $-\kappa_c d \log d$ allows diminishing returns and a finite overoptimization radius when $\kappa_c > 0$. This comparison family is deliberately low-dimensional and linear in its parameters after the feature transformation $\phi(d) = (1, d, -d \log d)$:

$$(\widehat{\gamma}_c, \widehat{\alpha}_c, \widehat{\kappa}_c) = \arg\min_{\gamma,\alpha,\kappa} \sum_{t=1}^{T_c} \left[G_{ct} - \gamma - d_{ct}(\alpha - \kappa \log d_{ct})\right]^2,$$

with the convention $d \log d = 0$ at $d = 0$. When $\kappa_c > 0$, the fitted stationary point is

$$d_c^* = \exp(\alpha_c/\kappa_c - 1), \qquad G_c(d_c^*) = \gamma_c + \kappa_c d_c^*.$$

Both fits are evaluated by RMSE and by whether the implied peak lies inside the observed KL range. We use these fitted peaks as diagnostics, not as extrapolated deployment recommendations.

Across the 16 local trajectories, the median RMSE is 63.3 Elo for the theory-implied quadratic and 63.5 Elo for the Gao-style comparison. The quadratic has three fitted peaks inside the observed KL range; the Gao-style form has six finite-radius fits but only two in-range peaks. This means the existing trajectories are useful for measuring local signal and curvature, but they mostly measure the rising part of the curve rather than a complete overoptimization frontier. The reproducibility artifacts under `repro_artifacts/scaling_laws/` include the script, per-configuration coefficients, and summary report.

A.5 Statistical Convergence of Preference Reward Estimators

This subsection formalizes what static reward-model training can and cannot guarantee. It is a statistical statement about preference estimation under a fixed data distribution, not a claim that nonconvex neural fine-tuning finds a global optimum. Let $z = (x, y_A, y_B)$ and $h \in \{+1, -1\}$ denote the preferred response, with $h = +1$ meaning $y_A \succ y_B$. A reward estimator is represented by a score function $f \in \mathcal{F}$, where $f(z) > 0$ predicts $y_A \succ y_B$. The logistic preference risk under a distribution P is

$$\mathcal{R}_P(f) = \mathbb{E}_{(z,h)\sim P}\left[\ell(hf(z))\right], \qquad \ell(u) = \log(1 + e^{-u}).$$

Given n iid preference examples from P, let $\widehat{f}$ be an approximate empirical risk minimizer satisfying

$$\widehat{\mathcal{R}}_n(\widehat{f}) \le \inf_{f\in\mathcal{F}} \widehat{\mathcal{R}}_n(f) + \varepsilon_{\mathrm{opt}}.$$

Theorem 2 (Static preference-estimator convergence)**.** *Assume $|f(z)| \le B$ for all $f \in \mathcal{F}$ and all z. Let $B_\ell = \log(1 + e^B)$ and let $\mathfrak{R}_n(\ell \circ \mathcal{F})$ denote the empirical Rademacher complexity of the labeled loss class $\{(z, h) \mapsto \log(1 + e^{-hf(z)}) : f \in \mathcal{F}\}$. With probability at least $1 - \delta$,*

$$\mathcal{R}_P(\widehat{f}) - \inf_{f\in\mathcal{F}} \mathcal{R}_P(f) \le 4\mathfrak{R}_n(\ell \circ \mathcal{F}) + 6B_\ell\sqrt{\frac{\log(4/\delta)}{2n}} + \varepsilon_{\mathrm{opt}}.$$

For fixed dimension d, an exact or sufficiently accurate consistent M-estimator with identifiable minimizer θ^* satisfies the following rates under standard smoothness, finite-score-variance and nonsingular-Hessian regularity conditions. Optimization error must be negligible at the relevant rate, in particular $o_p(n^{-1})$ for the displayed excess-risk rate:

$$\|\widehat{\theta} - \theta^*\| = O_p\left(\sqrt{\frac{d}{n}}\right), \qquad \mathcal{R}_P(f_{\widehat{\theta}}) - \mathcal{R}_P(f_{\theta^*}) = O_p\left(\frac{d}{n}\right).$$

Proof. The first inequality is the standard uniform-convergence bound for approximate ERM: symmetrization and concentration give

$$\sup_{f\in\mathcal{F}} \left|\mathcal{R}_P(f) - \widehat{\mathcal{R}}_n(f)\right| \le 2\mathfrak{R}_n(\ell \circ \mathcal{F}) + 3B_\ell\sqrt{\frac{\log(4/\delta)}{2n}}$$

with probability at least $1 - \delta$. Applying this bound to $\widehat{f}$ and a population-risk minimizer (or an approximating sequence), together with the approximate empirical-risk-minimization inequality, yields the displayed bound including $\varepsilon_{\mathrm{opt}}$. Under the stated fixed-dimensional regularity conditions, a score expansion around the consistent minimizer gives parameter error $O_p(\sqrt{d/n})$ and a second-order risk expansion gives $O_p(d/n)$ excess risk. □

The theorem is a fixed-distribution estimation statement, not a guarantee that increasing neural model size or training time improves accuracy. Logistic calibration bounds excess classification risk relative to the Bayes rule when approximation error is included; small in-class excess risk alone does not guarantee negligible disagreement with Gold. Let P_0 be the static distribution, Q_t the policy-induced pair distribution and $\eta_{\widehat{f}}(z)$ the proxy–Gold disagreement indicator. Even when assumptions give control of $\mathbb{E}_{P_0}[\eta_{\widehat{f}}]$, online DPO depends on $\mathbb{E}_{Q_t}[\eta_{\widehat{f}}]$ and gradient-weighted variants.

Coverage is the missing condition. If $Q_t \ll P_0$ with density ratio $r_t = dQ_t/dP_0$, then

$$\mathbb{E}_{Q_t}[\eta_{\widehat{f}}] = \mathbb{E}_{P_0}[r_t(z)\eta_{\widehat{f}}(z)] \le \sqrt{\mathbb{E}_{P_0}[\eta_{\widehat{f}}(z)]\,(1 + \chi^2(Q_t\|P_0))}.$$

If the stronger bounded-coverage condition $r_t(z) \le C$ holds, then

$$\mathbb{E}_{Q_t}[\eta_{\widehat{f}}] \le C\,\mathbb{E}_{P_0}[\eta_{\widehat{f}}].$$

These bounds provide sufficient conditions for transfer when the policy-induced pair distribution is adequately covered by the static distribution. Coverage is not necessary in every special case, and these assumptions are not established by static accuracy alone. Online DPO changes the response distribution; static estimators can consequently be worse rewarders when errors concentrate on policy-induced pairs.

A.6 Why Static Accuracy Does Not Control Online DPO Updates

This subsection formalizes the update-level difference between a GenRM used as a static evaluator and the same GenRM used as an online reward signal. The analysis is local: it studies the expected DPO gradient at a fixed online step t and does not claim convergence for the full non-stationary training process. Its purpose is to identify the quantities that static validation accuracy fails to measure.

Let $z = (x, y^+, y^-)$ denote a pair oriented by the Gold evaluator V_{gold}, so y^+ is Gold-preferred to y^-. Let Q_t be the distribution over such Gold-oriented pairs induced by sampling $x \sim \mathcal{P}_{\text{train}}$ and $y_1, y_2 \sim \pi_{\theta_t}(\cdot \mid x)$ at online step t. Define the DPO margin

$$\Delta_\theta(z) = \log \frac{\pi_\theta(y^+ \mid x)}{\pi_{\text{ref}}(y^+ \mid x)} - \log \frac{\pi_\theta(y^- \mid x)}{\pi_{\text{ref}}(y^- \mid x)}.$$

The Gold-label and flipped-label pair losses are

$$\ell_+(\Delta) = -\log \sigma(\beta\Delta), \qquad \ell_-(\Delta) = -\log \sigma(-\beta\Delta).$$

Proposition 2 (Label flips induce a structured DPO gradient bias)**.** *For any differentiable policy π_θ and any pair z,*

$$\nabla_\theta \ell_-(\Delta_\theta(z)) - \nabla_\theta \ell_+(\Delta_\theta(z)) = \beta \nabla_\theta \Delta_\theta(z).$$

Proof. For $s \in \{+1, -1\}$,

$$\frac{d}{d\Delta}\{-\log \sigma(s\beta\Delta)\} = -s\beta\sigma(-s\beta\Delta).$$

Thus

$$\nabla_\theta \ell_+(\Delta_\theta) = -\beta\sigma(-\beta\Delta_\theta)\nabla_\theta\Delta_\theta, \qquad \nabla_\theta \ell_-(\Delta_\theta) = \beta\sigma(\beta\Delta_\theta)\nabla_\theta\Delta_\theta.$$

Subtracting and using $\sigma(u) + \sigma(-u) = 1$ gives the result. □

Let $\eta_v(z) = \Pr[v(z) \ne V_{\text{gold}}(z) \mid z]$ be the proxy's conditional disagreement probability on a Gold-oriented pair. If the proxy and Gold update on the same pair distribution Q_t, then Proposition 2 gives

$$G_t^v - G_t^{\text{gold}} = \beta \mathbb{E}_{z\sim Q_t}\left[\eta_v(z)\nabla_\theta\Delta_{\theta_t}(z)\right],$$

where G_t^v and G_t^{gold} denote the expected proxy and Gold DPO gradients at θ_t. Consequently,

$$\|G_t^v - G_t^{\text{gold}}\| \le \beta \mathbb{E}_{z\sim Q_t}\left[\eta_v(z)\|\nabla_\theta\Delta_{\theta_t}(z)\|\right].$$

The natural online error diagnostic is therefore the gradient-weighted disagreement

$$\mathcal{E}_t(v) = \mathbb{E}_{z\sim Q_t}\left[\eta_v(z)\|\nabla_\theta\Delta_{\theta_t}(z)\|\right],$$

not the static error $\mathbb{E}_{z\sim P_0}[\eta_v(z)]$ under the fixed validation-pair distribution P_0.

This distinction also clarifies why modest average disagreement can still be damaging. If a proxy flips a fixed pair with probability η, the expected pairwise DPO loss is

$$L_\eta(\Delta) = (1-\eta)\ell_+(\Delta) + \eta\ell_-(\Delta).$$

For $0 < \eta < 1$, the unconstrained optimum satisfies

$$\sigma(\beta\Delta^*) = 1-\eta, \qquad \Delta^* = \frac{1}{\beta}\log\frac{1-\eta}{\eta}.$$

Thus $\eta < 1/2$ still favors the Gold response but only up to a finite margin, $\eta = 1/2$ carries no directional signal, and $\eta > 1/2$ pushes the margin against Gold. Online DPO is especially sensitive to errors on the policy-induced pairs for which the proxy crosses this local threshold.

Static accuracy controls this online quantity only under distributional assumptions. If P_0 is the static validation distribution and Q_t is the online pair distribution, then

$$\mathbb{E}_{Q_t}[\eta_v] - \mathbb{E}_{P_0}[\eta_v] = \int \eta_v(z)\, d(Q_t - P_0)(z).$$

With $\mathrm{TV}(P,Q) = \sup_A |P(A) - Q(A)|$ and $0 \le \eta_v \le 1$,

$$|\mathbb{E}_{Q_t}[\eta_v] - \mathbb{E}_{P_0}[\eta_v]| \le \mathrm{TV}(Q_t, P_0).$$

Equivalently, if $Q_t \ll P_0$ and $r_t = dQ_t/dP_0$, then

$$\mathbb{E}_{Q_t}[\eta_v] = \mathbb{E}_{P_0}[r_t(z)\eta_v(z)] \le \sqrt{\mathbb{E}_{P_0}[\eta_v(z)]\,\mathbb{E}_{P_0}[r_t(z)^2]},$$

where the last inequality uses Cauchy–Schwarz and $\eta_v(z)^2 \le \eta_v(z)$. Since $\mathbb{E}_{P_0}[r_t^2] = 1 + \chi^2(Q_t\|P_0)$, low static error is informative only when the online pair distribution has not shifted too far or when errors do not concentrate in high-density-ratio regions.

The actual online-DPO trainer has one additional complication: it only retains pairs for which the proxy produces a usable, position-consistent preference. Let $S_v(z) \in \{0,1\}$ be this retention event, $a_v(z) = \Pr(S_v(z) = 1 \mid z)$, and $\rho_v = \mathbb{E}_{Q_t}[a_v(z)]$. When $\rho_v > 0$, the proxy-retained distribution is

$$Q_t^v(z) = \frac{a_v(z)}{\rho_v} Q_t(z).$$

Let $g_t^{\mathrm{gold}}(z) = \nabla_\theta \ell_+(\Delta_{\theta_t}(z))$ and $d_t(z) = \nabla_\theta \Delta_{\theta_t}(z)$. Let $\eta_v^S(z)$ be the probability that proxy v gives the wrong preference conditional on retaining z.

Proposition 3 (Proxy filtering separates selection bias from label bias)**.** *Under the definitions above,*

$$G_t^v - G_t^{\mathrm{gold}} = \underbrace{\left(\mathbb{E}_{z\sim Q_t^v}[g_t^{\mathrm{gold}}(z)] - \mathbb{E}_{z\sim Q_t}[g_t^{\mathrm{gold}}(z)]\right)}_{\text{selection bias}} + \underbrace{\beta\mathbb{E}_{z\sim Q_t^v}\left[\eta_v^S(z) d_t(z)\right]}_{\text{label bias}}.$$

Proof. Write the expected proxy update after filtering as

$$G_t^v = \mathbb{E}_{z\sim Q_t^v}\left[(1-\eta_v^S(z))g_t^{\mathrm{gold}}(z) + \eta_v^S(z)\nabla_\theta\ell_-(\Delta_{\theta_t}(z))\right].$$

Add and subtract $\mathbb{E}_{Q_t^v}[g_t^{\mathrm{gold}}(z)]$, and then apply Proposition 2 inside the remaining expectation. □

If $\|g_t^{\text{gold}}(z)\| \le M_t$ and $\|d_t(z)\| \le W_t$, this decomposition implies the crude but useful bound

$$\|G_t^v - G_t^{\text{gold}}\| \le 2M_t \ \mathrm{TV}(Q_t^v, Q_t) + \beta W_t \, \mathbb{E}_{z\sim Q_t^v}[\eta_v^S(z)].$$

The first term is the cost of proxy-dependent retention; the second is the cost of wrong labels on retained pairs.

Finally, freeze the pair distribution at step t and define the local Gold-DPO objective

$$\mathcal{L}_t^{\text{gold}}(\theta) = \mathbb{E}_{z\sim Q_t}[\ell_+(\Delta_\theta(z))].$$

If $\mathcal{L}_t^{\text{gold}}$ is L-smooth near θ_t and a proxy step uses $\theta_{t+1} = \theta_t - \alpha G_t^v$, then

$$\mathcal{L}_t^{\text{gold}}(\theta_{t+1}) \le \mathcal{L}_t^{\text{gold}}(\theta_t) - \alpha\langle G_t^{\text{gold}}, G_t^v\rangle + \frac{L\alpha^2}{2}\|G_t^v\|^2.$$

Thus the first-order condition for a proxy step to remain Gold-improving is $\langle G_t^{\text{gold}}, G_t^v\rangle > 0$. If $G_t^v = G_t^{\text{gold}} + \delta_t^v$, this becomes $\langle G_t^{\text{gold}}, \delta_t^v\rangle > -\|G_t^{\text{gold}}\|^2$. The decomposition above identifies the two ways this condition can fail: proxy filtering can change the retained-pair distribution, and proxy label errors can add high-gradient update components anti-aligned with the Gold gradient.

A positive inner product is a descent-direction condition, not a guarantee for an arbitrary finite update. The smoothness bound guarantees improvement for sufficiently small positive α, with the step remaining in the smooth neighborhood. Static evaluator accuracy measures neither the retained-pair distribution Q_t^v nor gradient-weighted disagreement. The stored artifacts do not contain the per-pair margins and gradient features needed to estimate this decomposition directly.

A.7 CPU-Only Robustness Diagnostics

After constructing the main figures, we analyzed stored artifacts without new model sampling. Table 3 uses independent winner/loser-row bootstrap resampling and regularized refits. An auxiliary agreement analysis matched unordered checkpoint-player pairs in row order, without unique prompt/response identifiers; dropped records can break that alignment, so its percentages are not treated as verified per-response agreement. Generation surface diagnostics use stored peak-Gold validation outputs and measure response length, duplicate-line rate, distinct 4-gram ratio and first-line concentration. These are descriptive style indicators, not preference labels. The analysis files are retained in the research archive; this source upload does not imply public availability of the private experiment repository or complete raw data.

A.8 Follow-Up Audit with a Different Learned Judge

We report an existing follow-up diagnostic separately from the main Gold arena. For 4B policies trained with 4B and 14B rewarders, we selected each mode's peak-Gold snapshot and sampled 40 validation prompts per setting with seed 17. Response presentation order was randomized. The judge, `Qwen3-235B-A22B-Instruct-2507`, received the prompt and two responses without their training-mode labels. It judged prompt-following, coherence, specificity, originality and writing quality, with an instruction not to favor length. It produced one Answer-Only verdict per pair at temperature zero with a 24-token limit; no swapped-order repeat was performed. All 80 verdicts were parseable.

The diagnostic is inconclusive overall. Its two settings, selected checkpoints, changed rubric, shared Qwen3 family and single verdict per pair limit interpretation. It neither reproduces a human evaluation nor supplies a thinking-style Gold control; it makes the evaluator dependence of the main result explicit.

Rewarder size (4B policy)	Answer-Only wins	Comparisons	95% Wilson interval
4B	17	40	28.5–57.8%
14B	25	40	47.0–75.8%
Combined	42	80	41.7–63.1%

Table 6: A different learned judge does not establish an overall preference advantage in this limited diagnostic. These descriptive binomial intervals are not prompt-cluster or retraining intervals; the 80 comparisons contain 76 unique prompts, with four prompts shared across settings.

B Prompt Templates

We provide the exact prompt templates used for policy generation and GenRM pairwise evaluation. All models receive these prompts without modification.

B.1 Policy Prompt

The policy model receives the following prompt template for creative writing generation:

```
Creative Writing Prompt:
{QUESTION}
Goal: write an original short story that directly answers the prompt.
Requirements: Output only the story text. no titles, notes, commentary
or meta-text.
```

The `QUESTION` placeholder is replaced with the specific creative writing instruction from the evaluation set.

B.2 GenRM Pairwise Evaluation Prompt

The generative reward model receives the following prompt template for pairwise comparison:

```
Two AI models were given the same instruction and each produced a reply.
Your task is to judge which reply better fulfills the instruction.

[BEGIN INSTRUCTIONS GIVEN TO BOTH MODELS]
{POLICY_PROMPT}
[END INSTRUCTIONS]

[BEGIN REPLY A]
{RESPONSE_A}
[END REPLY A]

[BEGIN REPLY B]
{RESPONSE_B}
[END REPLY B]

Output ONLY one of:
<answer>[[A]]</answer>
<answer>[[B]]</answer>
```

The placeholders are filled with the full policy prompt and the two candidate responses. The model outputs a structured verdict indicating which response better satisfies the original instruction.